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1. Introduction to Association Rule Mining - Market Baset Analysis (MBA)

- Association rules are **if-then** statements that show **the probability of relationships** between data items within large data sets in various types of databases.
- At a basic level, association rule mining involves the use of machine learning models to analyze data for patterns, called **co-occurrences**, in a database.
- It identifies **frequent if-then associations**, which themselves are the association rules.
- For example, if 75% of people who buy cereal also buy milk, then there is an evidence for pattern in the transactional data that customers who buy cereal often buy milk.

For example, the rule:

$$\{\text{cereal}\} \Rightarrow \{\text{milk}\}$$

Such information can be used as the basis for decisions about marketing activities such as, e.g., promotional pricing or product placements.

Market Basket Analysis
Are Bread Purchased When Banana And Egg Are Bought Together?
Where Should Apple Juice Be Placed In The Store To Maximize Their Sales?
Is Soda Typically Purchased With Bananas?
Does The Brand Of Soda Make A Difference?
How Are The Demographics Of The Neighborhood Affecting What Customers Are Buying?

للاستفسار فرعي 108 - 06/5814168

رقم الفاتو 62465 الرقم الضريبي :16650476

كاش 6

التاريخ28/05/2025 الساعه : 3:35:00 pm

ف المادة	الكمية	السعر	المجموع
نبيل دجاج تندر 900 غم	1	3.750	3.750
نبيل اصابع سمك مجمد 40	2	2.390	4.780
خضار مشكلة السنبلة 450	2	1.600	3.200
السنبلة بامية اكسترا 400	2	2.590	5.180
سبانخ السنبلة 400 غم	2	0.900	1.800
جبنة فاكيوم روابي جرش (	1	3.550	3.550
مكدوس بلدي التميمي 1 كغ	1	3.650	3.650
برداييس خبز طابون *4	1	0.600	0.600
غيث خبز مسخن *5	1	0.850	0.850
مخلل هلابينو	0.494	1.990	0.983
ديلايت قشقوان	0.3	8.190	2.457
مشمش حموي سوبر	0.796	1.750	1.393
سمسمية مصرية	0.816	1.750	1.428
مفروم عجل + خاروف	0.335	8.490	2.844
مفروم عجل + خاروف	0.355	8.490	3.014
دجاج تندوري مقطع متبل لا	0.98	3.250	3.185

المجموع النهائي: 42.664
فيزا : 42.664
المدفوع 0.000
الباقي 0.000
عدد المواد : 20

شكرا لزيارتكم
لا يتم تبديل البضاعة الا بوجود
فاتورة شراء خلال 24 ساعة

للاستفسار فرعي 108 - 06/5814168

رقم الفاتو 62464 الرقم الضريبي :16650476

كاش 6

التاريخ28/05/2025 الساعه : 3:34:00 pm

ف المادة	الكمية	السعر	المجموع
ارم اند همر الفالس	1	1.350	1.350
ارم اند همر معجون اسنان	1	1.350	1.350
معجون اورل بي 123	1	1.390	1.390
المراعي لبنة 500 غم	1	2.250	2.250
صابون حمام نابلسي حبة و	1	0.550	0.550
عصير الربيع اناناس 1 لتر	1	1.400	1.400
عصير الربيع كوكتيل 1 لتر	1	1.400	1.400
هاربيك سيترس 495 مل	1	1.200	1.200
هاربيك لافندر 495 مل	1	1.200	1.200
كفوف لاتيكس اسود وسط*3	1	2.490	2.490
تشيروز عسل 375 غم	1	3.750	3.750
رايس كيك طبيعي 100 غم	2	0.950	1.900
رايس كيك سمسم 100 غم	1	0.950	0.950
غاستو مناديل مبللة 96	1	0.990	0.990
سانينو معجون اسنان الحسـ	1	0.990	0.990
جبنة برمزان ايطالي	0.166	12.990	2.156
جبنة برمزان ايطالي	0.354	12.990	4.598
فلفل قرن الغزال	0.552	0.650	0.359
بندورة عناقيد كبيرة	0.7	1.100	0.770
مخلل هلابينو	0.284	1.990	0.565
خيار	0.44	0.790	0.348

المجموع النهائي: 31.956
فيزا : 31.956
المدفوع 0.000
الباقي 0.000
عدد المواد : 22

شكرا لزيارتكم
لا يتم تبديل البضاعة الا بوجود
فاتورة شراء خلال 24 ساعة

The Apriori Algorithm

Apriori is the most common algorithm used for MBA. It identifies frequent itemsets and builds association rules from them by leveraging the principle that **subsets of frequent itemsets must also be frequent.**

Key Notes:

- Apriori is an algorithm for identifying frequent item-sets and association rule learning over relational databases.
- It was proposed by Agrawal and Srikant in 1994.
- Apriori is designed to operate on databases containing transactions (for example, collections of items bought by customers, or details of a website frequentation or IP addresses).
- It proceeds by identifying the frequent individual items in the database and extending them to larger and larger item sets as long as those item sets appear sufficiently often in the database.
- The frequent item sets determined by Apriori can be used to determine association rules which highlight general trends in the database.
- This has applications in domains such as market basket analysis.
- Apriori uses a "bottom up" approach, where frequent subsets are extended one item at a time (a step known as candidate generation), and groups of candidates are tested against the data.
- The algorithm terminates when no further successful extensions are found.

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Key Concepts:

1. **Association Rules**:

- These are if-then rules that identify relationships between items. For example, "If a customer buys bread, they are likely to buy butter."

2. **Metrics**:

2.1. **Support**:

Support is a measure that represents the frequency or proportion of transactions in the dataset that contain a given item set or pattern. In other words, it measures how often a particular combination of items appears together in the dataset. Support is a **Probability Based** measure that has a value between 0 and 1. $$ \text{Support(X)} = \frac{\text{Number of transactions containing X}}{\text{Total transactions}} $$

Valid Values for Support:

- **Range**: The value of support ranges from **0% to 100%**:
- **0%**: The itemset does not appear in any transaction.

- **100%**: The itemset appears in all transactions.
- *Interpreting Support:**
- **High support**: The itemset is common and may represent a strong association.
- **Low support**: The itemset is rare and might not provide actionable insights.

Thresholds in Practice

- In practice, a minimum support threshold is specified to filter out infrequent itemsets, ensuring only itemsets that appear frequently in the dataset are considered for further analysis.
- Generally speaking, the minimum values for **support**, **confidence**, and **lift** depend on the specific dataset and the business context, but here are some common guidelines and starting points:
- Setting a very high support threshold can exclude less frequent but potentially interesting itemsets.
- If support is too low, the rule may represent rare occurrences, which might not be meaningful or actionable.
    - **Typical Threshold**:
- **5-10%**: Frequently used in retail datasets where only a small fraction of products are bought together.
- **Higher Thresholds (20-30%)**: For smaller datasets or when focusing on very popular combinations.

Example:
If only 2 out of 10,000 transactions involve a rule, it's unlikely to be significant. Start with **5-10%** to ensure relevance.

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2.2. Confidence:

Confidence in Apriori is a measure that calculates the **probability** of the rule being true, in other words, it measures the reliability of the rule. Having higher confidence means that when the antecedent occurs, the consequent is likely to follow.

$$\text{Confidence}(\text{X} \rightarrow \text{Y}) = \frac{\text{Support}(\text{X} \cap \text{Y})}{\text{Support}(\text{X})}$$

Valid Values for Confidence

- **Range**: Confidence values range from **0 to 1**:

- **0**: The rule $X \to Y$ never holds true.
- **1**: The rule $X \to Y$ always holds true (perfect confidence).

Interpreting Confidence

1. **High Confidence (≈ 1)**:
   - Indicates that Y almost always occurs when X occurs.
   - Example: If $X = \{\text{bread}\}$ and $Y = \{\text{butter}\}$, a high confidence
     $$Confidence(X \to Y)$$
     implies that customers buying bread are very likely to buy butter.
2. **Low Confidence (≈ 0)**:
   - Indicates that Y rarely occurs when X occurs.
   - Example: If $X = \{\text{bread}\}$ and $Y = \{\text{cereal}\}$, a low confidence implies that buying bread is not strongly associated with buying cereal.

Thresholds in Practice

The minimum values for **confidence** depend on the specific dataset and the business context, but here are some common guidelines and starting points:

- **Typical Threshold**:
  - **70-80%**: Commonly used as a starting point for reliable rules.
  - Adjust based on your use case:
    - Lower threshold (50-60%) for exploratory insights.
    - Higher threshold (90%+) for high-accuracy recommendations.
- **Reason**: Rules with low confidence may not reliably predict the consequent item, making them less actionable.

Example:
A rule with 30% confidence means that only 30% of the time, the consequent is bought when the antecedent is bought. This may not justify a marketing strategy.

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Confidence vs. Other Metrics

- Having **High Confidence** results can lead to misleading rules if Y is very frequent in the dataset. For example, a high-confidence rule might occur simply because Y is common, not because X and Y are strongly related.
- To address this, metrics like **lift** or **conviction** are often used alongside confidence.

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2.3. Lift:

Lift measures the **ratio** or the strength of the association between two items or itemsets. Specifically, it compares the observed co-occurrence of items to their expected co-occurrence if they were independent. In other words, lift shows whether the items are occurring together more often than expected by chance. Higher values are better.

$$\text{Lift}(\text{X} \rightarrow \text{Y}) = \frac{\text{Confidence}(\text{X} \rightarrow \text{Y})}{\text{Support}(\text{Y})}$$

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Support}(X \cap Y)}{\text{Support}(X) \times \text{Support}(Y)}$$

Valid Values for Lift

1. **Greater than 1**:
   - Indicates a **positive association** between X and Y.
   - X and Y are more likely to occur together than if they were independent.
2. **Equal to 1**:
   - Indicates **no association** between X and Y.
   - X and Y occur together just as often as would be expected under independence.
3. **Less than 1**:
   - Indicates a **negative association** between X and Y.
   - X and Y are less likely to occur together than if they were independent.

Thresholds in Practice

The minimum values for **lift** depend on the specific dataset and the business context, but here are some common guidelines and starting points:

- **Typical Threshold**:
  - **Lift > 1**: Indicates a positive association between items (start considering rules here).
  - **Lift > 3**: Strong and actionable association.
  - **Lift = 1**: No association (ignore these rules).
  - **Lift < 1**: Negative association (discard unless relevant for avoiding recommendations).

Example:
If Lift = 1.5, buying the antecedent increases the likelihood of buying the consequent by 50%. If Lift = 4, it's a much stronger association.

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Suggested Starting Points for General Use:

- **Support ≥ 5-10%**
- **Confidence ≥ 70-80%**
- **Lift > 1 (preferably > 3 for strong rules)**

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Rule Prioritization

To prioritize rules, combine the metrics:

1. **High Lift + High Confidence + Acceptable Support**: These are the most actionable rules.
   - Example: If $\text{Lift} = 4$, $\text{Confidence} = 85\%$, and $\text{Support} = 12\%$, this rule is strong and impacts many customers.
2. **Trade-offs**:
   - Rules with **low support** but **high confidence and lift** might still be valuable for niche promotions.
   - Rules with **high support** but **low lift** may indicate popular combinations that are obvious and less insightful.

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3. Step-by-Step Implementation for the Apriori Algorithm

We will go step-by-step through how the **Apriori algorithm** would process this dataset. We'll compute frequent itemsets and generate rules using the given data, along with thresholds for **support**, **confidence**, and **lift**.

Let's assume a small dataset with 12 transactions:

Transaction ID	Item 1	Item 2	Item 3	Item 4	Item 5
1	Milk	Egg	Bread	Butter	
2	Milk	Butter	Egg	Ketchup	
3	Bread	Butter	Ketchup		
4	Milk	Bread	Butter		

Transaction ID	Item 1	Item 2	Item 3	Item 4	Item 5
5	Bread	Butter	Cookies		
6	Milk	Bread	Butter	Cookies	
7	Milk	Cookies			
8	Milk	Bread	Butter		
9	Bread	Butter	Egg	Cookies	
10	Milk	Butter	Bread		
11	Milk	Bread	Butter		
12	Milk	Bread	Cookies	Ketchup	

We set out thresholds as following:

- **Minimum Support**: 50% (focus on items present in at least half of the transactions), this value can be set to 5% or 10% in real-life situations with large datasets.
- **Minimum Confidence**: 70%.
- **Lift**: Lift > 1 for actionable rules.

Step 1: Dataset Transformation Convert the dataset into a binary format:

Transaction	Milk	Egg	Bread	Butter	Ketchup	Cookies
1	1	1	1	1	0	0
2	1	1	0	1	1	0
3	0	0	1	1	1	0
4	1	0	1	1	0	0
5	0	0	1	1	0	1
6	1	0	1	1	0	1
7	1	0	0	0	0	1
8	1	0	1	1	0	0
9	0	1	1	1	0	1
10	1	0	1	1	0	0
11	1	0	1	1	0	0
12	1	0	1	0	1	1

Step 2: Identify Frequent 1-Itemsets The algorithm starts by calculating the support for each single item.

Item	Support Count	Support (%)	Frequent?
Milk	9	75%	Yes
Egg	3	25%	No
Bread	10	83%	Yes
Butter	10	83%	Yes
Ketchup	3	25%	No

Item	Support Count	Support (%)	Frequent?
Cookies	5	42%	No

Frequent 1-itemsets: `{Milk}`, `{Bread}`, `{Butter}`.

Step 3: Generate 2-Itemsets Now, combine frequent 1-itemsets into 2-itemsets and calculate their support.

Itemset	Support Count	Support (%)	Frequent?
{Milk, Bread}	7	58%	Yes
{Milk, Butter}	7	58%	Yes
{Bread, Butter}	9	75%	Yes

Frequent 2-itemsets: `{Milk, Bread}`, `{Milk, Butter}`, `{Bread, Butter}`.

Step 4: Generate 3-Itemsets Combine frequent 2-itemsets into 3-itemsets and calculate their support.

Itemset	Support Count	Support (%)	Frequent?
{Milk, Bread, Butter}	6	50%	Yes

Frequent 3-itemset: `{Milk, Bread, Butter}`.

Step 5: No Larger Itemsets Since there are no frequent 4-itemsets (support would drop below 50%), we stop here.

Step 6: Generate Association Rules

Now, use the frequent itemsets to generate association rules and calculate their metrics: **support**, **confidence**, and **lift**.

Rules from `{Milk, Bread, Butter}`:

1. **Rule**:

$$\text{Milk, Bread} \rightarrow \text{Butter}$$

   - **Support**:

$$P(\text{Milk, Bread, Butter}) = 6/12 = 50\%$$

   - **Confidence**:

$$P(\text{Butter}|\text{Milk, Bread}) = \frac{P(\text{Milk, Bread, Butter})}{P(\text{Milk, Bread})} = \frac{6}{7} = 85$$

   - **Lift**:

$$\text{Lift} = \frac{P(\text{Butter}|\text{Milk, Bread})}{P(\text{Butter})} = \frac{0.85}{0.83} \approx 1.02$$

2. **Rule**:

$$\text{Milk, Butter} \rightarrow \text{Bread}$$

   - **Support**:

$$P(\text{Milk, Bread, Butter}) = 6/12 = 50\%$$

   - **Confidence**:

$$P(\text{Bread}|\text{Milk, Butter}) = \frac{P(\text{Milk, Bread, Butter})}{P(\text{Milk, Butter})} = \frac{6}{7} = 85$$

   - **Lift**:

$$\text{Lift} = \frac{P(\text{Bread}|\text{Milk, Butter})}{P(\text{Bread})} = \frac{0.85}{0.83} \approx 1.02$$

3. **Rule**:

$$\text{Bread, Butter} \rightarrow \text{Milk}$$

   - **Support**:

$$P(\text{Milk, Bread, Butter}) = 6/12 = 50\%$$

   - **Confidence**:

$$P(\text{Milk}|\text{Bread, Butter}) = \frac{P(\text{Milk, Bread, Butter})}{P(\text{Bread, Butter})} = \frac{6}{9} \approx 66$$

   - **Lift**:

$$\text{Lift} = \frac{P(\text{Milk}|\text{Bread, Butter})}{P(\text{Milk})} = \frac{0.66}{0.75} \approx .88$$

Final Results

Frequent Itemsets:

- **1-itemsets**: `{Milk}, {Bread}, {Butter}`
- **2-itemsets**: `{Milk, Bread}, {Milk, Butter}, {Bread, Butter}`
- **3-itemset**: `{Milk, Bread, Butter}`

Rules: | Rule | Support | Confidence | Lift | Actionable? | |---|---|---|---|---|---|---|---|

$$\text{Milk, Bread} \rightarrow \text{Butter}$$

| 50% | 85% | 1.02 | Yes | |

$$\text{Milk, Butter} \rightarrow \text{Bread}$$

| 50% | 85% | 1.02 | Yes | |

$$\text{Bread, Butter} \rightarrow \text{Milk}$$

| 50% | 66% | .88 | No |

Interpretation and Actionable Insights | Rule | Explaniation | |---|---|---|---|

$$\text{Milk, Bread} \rightarrow \text{Butter}$$

|Customers buying Milk and Bread are highly likely (85%) to also buy Butter. Consider bundling these items. | |

$$\text{Milk, Butter} \rightarrow \text{Bread}$$

| Strong association; placing these items together could increase sales. | |

$$\text{Bread, Butter} \rightarrow \text{Milk}$$

| Suggests that Milk is a complementary product to Bread and Butter, Weak lift thought |

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3. **Python Code Example**

We will implement the **Apriori algorithm** using the `mlxtend` library for frequent itemset mining and rule generation.

a. Install Required Libraries Make sure you have the required libraries installed:

```
pip install pandas mlxtend
```

b. Load the Dataset We will represent the dataset as a binary **transaction matrix** where each row is a transaction, and each column is an item (1 = purchased, 0 = not purchased).

```
import pandas as pd

# Define the dataset
data = {
    "Milk":    [1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1],
    "Egg":     [1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0],
    "Bread":   [1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1],
    "Butter":  [1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0],
    "Ketchup": [0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1],
    "Cookies": [0, 0, 0, 0, 1, 1, 1, 0, 1, 0, 0, 1],
}

# Create a DataFrame
df = pd.DataFrame(data)
print("Transaction Dataset:")
df
```

```
Transaction Dataset:
```

	Milk	Egg	Bread	Butter	Ketchup	Cookies
0	1	1	1	1	0	0
1	1	1	0	1	1	0
2	0	0	1	1	1	0
3	1	0	1	1	0	0
4	0	0	1	1	0	1
5	1	0	1	1	0	1
6	1	0	0	0	0	1
7	1	0	1	1	0	0
8	0	1	1	1	0	1
9	1	0	1	1	0	0
10	1	0	1	1	0	0
11	1	0	1	0	1	1

Next steps: Generate code with df | View recommended plots | New interactive sheet

c. Apply the Apriori Algorithm We will use the `mlxtend` library to find frequent itemsets and generate association rules.

```
! pip install mlxtend
```

```
Requirement already satisfied: mlxtend in /usr/local/lib/python3.11/dist-packages (0.23.
Requirement already satisfied: scipy>=1.2.1 in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: numpy>=1.16.2 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: pandas>=0.24.2 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: scikit-learn>=1.3.1 in /usr/local/lib/python3.11/dist-pac
Requirement already satisfied: matplotlib>=3.0.0 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: joblib>=0.13.2 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packa
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.11/dist-package
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.11/dist-packages (fro
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packag
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/dist-pa
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from
```

```
from mlxtend.frequent_patterns import apriori, association_rules

# Step 1: Generate frequent itemsets with Apriori
frequent_itemsets = apriori(df, min_support=0.5, use_colnames=True)

# Display frequent itemsets
print("\nFrequent Itemsets:")
print(frequent_itemsets)
```

```
Frequent Itemsets:
    support              itemsets
0  0.750000                 (Milk)
1  0.833333                (Bread)
2  0.833333               (Butter)
3  0.583333          (Bread, Milk)
4  0.583333         (Butter, Milk)
5  0.750000        (Butter, Bread)
6  0.500000  (Butter, Bread, Milk)
/usr/local/lib/python3.11/dist-packages/mlxtend/frequent_patterns/fpcommon.py:161: Depre
  warnings.warn(
```

d. Generate Association Rules We generate rules based on **confidence** and calculate **lift**.

Generate rules based on confidence

```
# Step 2: Generate association rules based on confidence
rules = association_rules(frequent_itemsets, num_itemsets=len(df), metric="confidence",

# Display the rules
print("\nAssociation Rules:")
print(rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']])
```

```
Association Rules:
      antecedents consequents   support  confidence      lift
0         (Bread)      (Milk)  0.583333    0.700000  0.933333
1          (Milk)     (Bread)  0.583333    0.777778  0.933333
2        (Butter)      (Milk)  0.583333    0.700000  0.933333
3          (Milk)    (Butter)  0.583333    0.777778  0.933333
4        (Butter)     (Bread)  0.750000    0.900000  1.080000
5         (Bread)    (Butter)  0.750000    0.900000  1.080000
6  (Butter, Milk)     (Bread)  0.500000    0.857143  1.028571
7   (Bread, Milk)    (Butter)  0.500000    0.857143  1.028571
```

We can also generate rules based on lift

```
# Step 2: Generate association rules based on lift
rules = association_rules(frequent_itemsets, num_itemsets=len(df), metric="lift", min_th

# Display the rules
print("\nAssociation Rules:")
print(rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']])
```

```
Association Rules:
      antecedents     consequents  support  confidence      lift
0        (Butter)         (Bread)     0.75    0.900000  1.080000
1         (Bread)        (Butter)     0.75    0.900000  1.080000
2  (Butter, Milk)         (Bread)     0.50    0.857143  1.028571
3   (Bread, Milk)        (Butter)     0.50    0.857143  1.028571
4        (Butter)   (Bread, Milk)     0.50    0.600000  1.028571
5         (Bread)  (Butter, Milk)     0.50    0.600000  1.028571
```

e. Filter and Interpret Rules You can filter rules for high lift or specific itemsets.

```
# Filter rules with Lift > 1
filtered_rules = rules[rules['lift'] > 1]
print("\nFiltered Rules with Lift > 1:")
print(filtered_rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']])
```

```
Filtered Rules with Lift > 1:
      antecedents     consequents  support  confidence      lift
0        (Butter)         (Bread)     0.75    0.900000  1.080000
1         (Bread)        (Butter)     0.75    0.900000  1.080000
2  (Butter, Milk)         (Bread)     0.50    0.857143  1.028571
3   (Bread, Milk)        (Butter)     0.50    0.857143  1.028571
4        (Butter)   (Bread, Milk)     0.50    0.600000  1.028571
5         (Bread)  (Butter, Milk)     0.50    0.600000  1.028571
```

Interpretation

1. $$\text{Milk, Bread} \rightarrow \text{Butter}$$
   - Support = 58%, Confidence = 87.5%, Lift = 1.05.
   - Suggest bundling these items for promotions.
2. $$\text{Milk, Butter} \rightarrow \text{Bread}$$
   - Similar metrics as above, showing strong relationships.
3. $$\text{Bread, Butter} \rightarrow \text{Milk}$$
   - High confidence but slightly weaker lift, still actionable.

Two Items Rules

If desired, association rules at the 1-item values of X could be specified. Consider the rule: Milk → Bread.

Out of total 12 transactions Milk occurs 9 times while (Milk, Bread) occurs 7 times.

The rule has a support level of 7/12 (or 58 percent) and a confidence level of 7/9 (or 77 percent).

Thus, the next valid association rule is as follows:

Milk → Bread {S = 58%, C = 77%}.